Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/07/2012
PI Name:	Bhattacharya, Sharmila		
Project Title:	Gravity and oxidative stress responses in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Ames Research Center		
PI Address 1:	Building: N236, Room: 213		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	03/01/2010	**End Date:**	02/28/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:			
Flight Assignment:	NOTE: Extended to 2/28/2014 per A. Chu/ARC (Ed., 3/4/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marcu, Oana (SETI Institute)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space environment of microgravity and radiation imposes stress conditions that affect the cellular and molecular responses of organisms and dictate their survival. A very fast response to stress is the production of reactive oxygen species (ROS), followed by intracellular oxidation. Space flight is associated with increased oxidative damage after return to 1g, suggesting that microgravity and/or the transition between gravity fields triggers a response. We propose to use a model organism used previously for space flight, the fruit fly Drosophila melanogaster. Studies in both hyper- and microgravity have already established many of the physiological alterations, and work from our own laboratory suggests that oxidative stress is implicated in the molecular response under both conditions. We will use hypergravity as a test environment for gravity alterations, and to dissect out molecular and cellular responses as well as long-term physiological outcomes. The goal is to provide a complete picture of the oxidative stress pathway in hypergravity, and to determine and select the key molecular functions and physiological read-outs to be tested in microgravity. The aims are to quantify gravity-dependent cellular ROS production and to determine the key points of entry into the oxidative pathway by using mutants that show a physiologically altered response to hypergravity. This approach will give a picture of the stress pathway, starting from fast cellular changes triggered by sudden changes in the gravity vector, to tissue-specific gene and enzyme activity alterations. Understanding the stress pathway in hypergravity opens the venue into looking at the opposite transition in the gravity load, from 1 g to microgravity during space flight, and will help predict the cellular and physiological response to other stresses imposed by the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Some of the imaging tools developed to measure the behavioral changes for this project will be useful for a variety of behavioral studies in different organisms on Earth. The general issue of oxidative stress is also of great relevance to aging and stress on Earth, and the studies here therefore provide yet another paradigm in which to study and understand these molecular responses.
Task Progress:	The main goal of this research is to identify gene networks involved in the oxidative stress response in altered gravity, in the fruit fly Drosophila melanogaster. The specific objectives that were accomplished during this period were as follows: 1. we were able to quantify the behavioral changes due to hypergravity, in select mutants (ubiquitous or targeted to the brain through RNA interference); 2. we quantified some of the molecules that trigger the changes in behaviors after exposure to hypergravity: the ROS (reactive oxygen species) levels in the fly brain; 3. we are working to integrate the data into gene networks that are relevant to the gravity stress.
Bibliography Type:	Description: (Last Updated: 03/02/2021)
Abstracts for Journals and Proceedings	Bhattacharya S, Marcu O. "Spaceflight changes Innate Immune Responses in the Drosophila melanogaster Host." Invited talk. ELGRA (European Low Gravity Research Association) Biennial Symposium and General Assembly 'Gravity: from µ to x!', Antwerp, Belgium, Sept. 6-9, 2011. ELGRA conference proceedings, September 2011. , Sep-2011
Abstracts for Journals and Proceedings	Bhattacharya S, Marcu O, Lera M, Sanchez M, Fahlen T, Higgins L, Shmygelska A. "Immune System Changes Triggered by Spaceflight in the Drosophila melanogaster Host." Invited talk at opening plenary session. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 40. , Nov-2011
Abstracts for Journals and Proceedings	Chan K, Inan O, Marcu O, Bhattacharya S. "Evaluating Signals from a System for Monitoring Locomotor Activity of Drosophila melanogaster under Gravitational Stress." Presented at 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 81. , Nov-2011
Abstracts for Journals and Proceedings	Coffee E, Chen L, Prabhakar A, Schuh [correction of Such] T, Smith J, Bhattacharya S. "Development of ISS Payload to Examine Flight Patterns of Drosophila melanogaster under Long-term Microgravity Exposure." Presented at 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 77. , Nov-2011
Abstracts for Journals and Proceedings	Prabhakar A, Schuh [correction of Such] T, Coffee E, Chen L, Smith J, Bhattacharya S. "Experimental Testing of Fruit Fly Behavior in the NASA ARC Vertical Motion Simulator." Presented at 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 74. , Nov-2011
Abstracts for Journals and Proceedings	Rankin-Gee E, Badani H, Marcu O, Hosamani R, Bhattacharya S. "Effects of Hypergravity on the Oxidative Stress Pathway of D. melanogaster." Presented at 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 77-78. , Nov-2011
Articles in Peer-reviewed Journals	Inan OT, Marcu O, Sanchez ME, Bhattacharya S, Kovacs GT. "A portable system for monitoring the behavioral activity of Drosophila." J Neurosci Methods. 2011 Oct 30;202(1):45-52. Epub 2011 Sep 3. PubMed PMID: 21907735 , Oct-2011

Awards	Rankin-Gee E. "Elyse Rankin-Gee won Honorable Mention Student Award in animal research category for 'Effects of Hypergravity on the Oxidative Stress Pathway of D. melanogaster,' 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011." Nov-2011
Significant Media Coverage	NASA Tech Briefs. " 'A Miniaturized Video System for Monitoring Drosophila Behavior.' Article about work done by Sharmila Bhattacharya of Ames Research Center; Omer Inan, Gregory Kovacs, and Mozziyar Etemadi of Stanford University; Max Sanchez of Lockheed Martin; and Oana Marcu of National Space Grant Foundation." NASA Tech Briefs, 2011 Apr., Apr-2011